

1. STUDENT FEATURES & LOGIC

Core Purpose: To provide personalized, adaptive learning and clear visibility into exam readiness.

- **Secure Profile Access:**
 - *Feature:* PIN/Biometric login for shared devices.
 - *Logic:* Allows multiple students to safely use a single classroom or home tablet without cross-contaminating academic data.
 - **WAEC Readiness Dashboard:**
 - *Feature:* A real-time visual meter (0-100) displaying projected WAEC success.
 - *Logic:* Aggregates historical quiz scores and teacher diary inputs, weighting them against standard WAEC grading curves to output a live probability score.
 - **Diagnostic Hotspot Viewer:**
 - *Feature:* Highlights specific sub-subjects (e.g., "Quadratic Equations") marked in Red/Yellow/Green.
 - *Logic:* If a student scores < 65% on a topic assessment, the system locks it as a "Hotspot" on their dashboard until cleared by a passing CBT score.
 - **Offline CBT (Computer-Based Testing) Engine:**
 - *Feature:* Timed, adaptive mock exams and quizzes available offline.
 - *Logic:* Automatically pulls question sets targeting the student's active Hotspots. Question difficulty dynamically scales up or down based on the previous answer's correctness.
 - **24/7 AI WhatsApp Tutor:**
 - *Feature:* Conversational AI interface accessible via WhatsApp.
 - *Logic:* AI is strictly prompt-engineered to *never* give direct answers. It uses local analogies to explain concepts step-by-step and quizzes the student to verify understanding.
 - **Predictive Career Streaming (JS3):**
 - *Feature:* Automated end-of-year pathway recommendation (Science, Arts, or Commercial).
 - *Logic:* Algorithm evaluates 3 years of longitudinal performance and behavioral data to output a statistically backed career path recommendation, overriding guesswork.
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2. TEACHER FEATURES & LOGIC

Core Purpose: To eliminate administrative friction and provide instantaneous classroom analytics.

- **Offline-First & Device Binding:**
 - *Feature:* Phone number OTP login that locks to the specific device hardware.

- *Logic:* Prevents proxy check-ins. Teachers can work 100% offline; the app caches data locally and silently syncs when an internet connection is detected.
 - **Dynamic QR Morning Attendance:**
 - *Feature:* Mobile scanner to capture the school gate QR code.
 - *Logic:* App records the scan, tags it with the device's background GPS coordinates, and validates if the teacher is actually within the school compound before logging them as "Present."
 - **The 60-Second Digital Diary:**
 - *Feature:* A rapid data-entry interface for daily class logging.
 - *Logic Flow:*
 1. **Attendance:** Displays a "Present-by-default" student grid. Teacher only taps absentees.
 2. **Curriculum Selection:** Teacher taps pre-loaded WAEC topic chips covered in that lesson.
 3. **Readiness Slider:** Teacher drags a 0-100 slider to estimate general class comprehension.
 4. **Behavioral Pins:** Tap-scoring for class stamina, focus, and agility.
 5. **Commit:** Generates an immutable, UTC-timestamped record.
 - **Zero-Marking AI Homework Engine:**
 - *Feature:* Assessment generator and automatic grader.
 - *Logic:* Teacher inputs a topic, difficulty, and question count. The AI generates a standardized assignment. Once students complete it, the teacher uses the app's camera to scan the worksheet. The built-in OCR grades it instantly and updates student Hotspots.
 - **Missed Diary Nudges:**
 - *Feature:* Push notification alerts.
 - *Logic:* If the daily diary is not synced by 2:00 PM, the app triggers a local alarm/push notification prompting immediate completion.
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3. PARENT FEATURES & LOGIC

Core Purpose: To enforce accountability at home without requiring new app downloads or technical skills.

- **Automated Friday Progress Reports:**
 - *Feature:* Rich-text report cards sent directly to WhatsApp.
 - *Logic:* Every Friday at 16:00, the system compiles the student's weekly attendance, cleared/active Hotspots, and WAEC Readiness score into a formatted message and pushes it via the WhatsApp Business API.
- **SMS Fallback System:**
 - *Feature:* Plain-text message routing.

- *Logic:* The gateway checks if the parent's number has an active WhatsApp account. If it bounces or is inactive, the system automatically converts the report into a concise SMS text to guarantee delivery.
 - **Urgent Risk Alerts:**
 - *Feature:* Instant notification triggers.
 - *Logic:* If a student is marked absent for 3 consecutive days, or their WAEC readiness drops below 40%, the system overrides the Friday schedule and immediately sends a high-priority alert to the parent.
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4. SCHOOL ADMIN (PRINCIPAL) FEATURES & LOGIC

Core Purpose: To monitor campus-level compliance, teacher performance, and resource usage.

- **Campus Health Dashboard:**
 - *Feature:* Web interface showing daily school statistics.
 - *Logic:* Aggregates live data from teacher diaries to display total student attendance, teacher punctuality, and overall school curriculum velocity against the state calendar.
 - **Teacher Compliance Tracker:**
 - *Feature:* Roster tracking diary submissions.
 - *Logic:* Flags teachers who consistently miss the 2:00 PM diary sync. Acts as the Level 2 escalation point (Alerts principal at the 24-hour mark of a missed diary).
 - **SmartSpend Inventory Oversight:**
 - *Feature:* Digital ledger for school assets and expenses.
 - *Logic:* When textbooks or lab equipment arrive, the admin logs the inventory with a digital receipt. The system tracks allocations to specific classes/teachers, triggering alerts if assets are misreported or go missing.
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5. STATE & DISTRICT (MINISTRY) FEATURES & LOGIC

Core Purpose: To provide real-time, macroeconomic visibility and predictive policy control.

- **Statewide Governance Command Center:**
 - *Feature:* Macro web dashboard with drill-down capabilities.
 - *Logic:* Aggregates data from all districts. Allows the Permanent Secretary to view top-level WAEC readiness averages, and click down through District -> LGA -> School -> Specific Classroom.
- **Live District-Level Heatmaps:**
 - *Feature:* Geographic visualization of academic performance.

- *Logic:* Maps Lagos State. If multiple schools in a specific LGA (e.g., Alimosho) fail specific math nodes, that geographical zone glows "Red" on the map, allowing for targeted resource deployment.
 - **Automated Protocol Escalation Engine:**
 - *Feature:* Multi-tier accountability workflow.
 - *Logic:* * 0 Hours: Teacher misses diary.
 - 24 Hours: Alert routed to Principal's dashboard.
 - 48 Hours: Alert routed automatically to the District/LGA Supervisor's dashboard for direct intervention.
 - **Asset Tracking & Supplier Accountability:**
 - *Feature:* State-level view of the SmartSpend system.
 - *Logic:* Cross-references state procurement contracts against actual school-level inventory receipts. Flags discrepancies automatically to prevent resource leakage.
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6. CORE SYSTEM OPERATIONAL FLOW (The Engine Loop)

How the features interconnect (Diagnose → Fix → Test)

1. **DIAGNOSE (The Input):** The Teacher executes the 60-Second Digital Diary. The system ingests this data, cross-references it with recent student CBT scores, and calculates *Diagnostic Hotspots* and updates the *WAEC Readiness Meter*.
2. **FIX (The Intervention):** The system takes the updated Hotspots and does three things simultaneously:
 - Alerts the Teacher via their dashboard.
 - Generates a targeted AI Worksheet for the classroom.
 - Pushes a Friday WhatsApp report to the Parent detailing the exact topic to practice.
3. **TEST (The Validation):** The Student logs into the CBT Engine (or uses the WhatsApp Tutor). They are tested specifically on the Hotspot topic.
4. **RECALIBRATE:** The Student's score is processed. If they pass, the Hotspot turns Green, the WAEC Meter goes up, and the Teacher and Parent are notified of the victory. The loop restarts.